SINGAPORE SCHOOL

CONTINUAL ASSESSMENT 1

Secondary 1

Computer Science (Part 1) — ANSWER KEY

2026-27

Date: ___ / ___ / ___ **Name: _____**

Instructions for students:

- **Start answering the questions on time.**
- **Follow all the instructions carefully.**
- **Answer all questions.**
- **Revise**

CA 1 Marks (Part – 1)	**25**
CW	**05**
Total Marks Obtained	**30**

1. Choose the correct answer for each of the following multiple-choice questions. (1 x 10 = 10)

1. What is the base of the binary number system?

A) Base 2 B) Base 8
C) Base 10 D) Base 16

➔ Answer: A) Base 2

2. Convert the 4-bit binary number 1101 into decimal (denary):

A) 11 B) 12
C) 13 D) 14

➔ Answer: C) 13 (8 + 4 + 0 + 1 = 13)

3. Convert the decimal (denary) number 25 into an 8-bit binary number:

A) 00011001 B) 00010101
C) 00110001 D) 00011100

➔ Answer: A) 00011001 (16 + 8 + 1 = 25)

4. How many binary bits are represented by a single hexadecimal digit?

A) 2 bits B) 4 bits (1 nibble)
C) 8 bits (1 byte) D) 16 bits

➔ Answer: B) 4 bits (1 nibble)

5. In the hexadecimal number system, what decimal value does the letter 'E' represent?

A) 12 B) 13
C) 14 D) 15

➔ Answer: C) 14 (A=10, B=11, C=12, D=13, E=14, F=15)

6. Convert the binary number 1010 1111 into hexadecimal:

A) 9F B) AF
C) B7 D) AE

➔ Answer: B) AF (1010 = 10 = A, 1111 = 15 = F)

7. Convert the hexadecimal number 3C into an 8-bit binary number:

A) 0011 1100 B) 0011 1010
C) 0110 1100 D) 0011 1101

➔ Answer: A) 0011 1100 (3 = 0011, C = 12 = 1100)

8. In binary addition, what is the result of 1 + 1?

A) 2 with carry 0 B) 1 with carry 1
C) 0 with carry 1 D) 0 with carry 0

➔ Answer: C) 0 with carry 1 (1 + 1 = 10 in binary)

9. In binary addition, what is the result of adding three ones (1 + 1 + 1)?

A) 1 with carry 1 B) 0 with carry 1
C) 1 with carry 0 D) 3 with carry 0

➔ Answer: A) 1 with carry 1 (1 + 1 + 1 = 11 in binary)

10. What is the result of adding the 4-bit binary numbers 0101 + 0011?

A) 0111 B) 1000

C) 1001 D) 1010

➔ Answer: B) 1000 (5 + 3 = 8 = 1000 in binary)

2. Convert each of the following 8-bit binary numbers into decimal (denary). Show your working. **(1 x 3 = 3)**

(a) 0010 1101

Worked Solution:
32 + 8 + 4 + 1 = 45

(b) 0101 0110

Worked Solution:
64 + 16 + 4 + 2 = 86

(c) 1001 1011

Worked Solution:
128 + 16 + 8 + 2 + 1 = 155

3. Convert each of the following decimal (denary) numbers into 8-bit binary. Show your working. **(1 x 3 = 3)**

(a) 42

Worked Solution:
32 + 8 + 2 = 0010 1010

(b) 99

Worked Solution:
64 + 32 + 2 + 1 = 0110 0011

(c) 205

Worked Solution:
128 + 64 + 8 + 4 + 1 = 1100 1101

4. Convert each of the following 8-bit binary numbers into hexadecimal. **(1 x 3 = 3)**

(a) 0100 1101

Worked Solution:
0100 = 4, 1101 = 13 (D) --> 4D

(b) 1011 0111

Worked Solution:
1011 = 11 (B), 0111 = 7 --> B7

(c) 1110 1010

Worked Solution:
1110 = 14 (E), 1010 = 10 (A) --> EA

5. Convert each of the following hexadecimal values into 8-bit binary. **(1 x 3 = 3)**

(a) 59

Worked Solution:
5 = 0101, 9 = 1001 --> 0101 1001

(b) A6

Worked Solution:
A (10) = 1010, 6 = 0110 --> 1010 0110

(c) F8

Worked Solution:
F (15) = 1111, 8 = 1000 --> 1111 1000

6. Perform binary addition on each of the following pairs of 8-bit binary numbers. **(1.5 x 2 = 3)**
Show all carry bits clearly.

(a) 0010 0111 + 0011 1010

Worked Solution:
0010 0111 (39) + 0011 1010 (58) = 0110 0001 (97) [Carry row: 0011 1100]

(b) 0101 1101 + 0010 1110

Worked Solution:
0101 1101 (93) + 0010 1110 (46) = 1000 1011 (139) [Carry row: 0111 1000]